Estimating ancestry composition from GWAS summary statistics

Frequency projection and two LD-moment channels

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Abstract

A GWAS summary-statistics file does not uniquely determine an ancestry composition. We distinguish three working estimands: the simplex projection of reported effect-allele frequencies (EAF) onto a fixed reference-frequency panel; linear coefficients in a model for the covariance of association z-statistics; and a rank-one bilinear LD-score model for χ^2 statistics. Only under additional sampling assumptions do any of these coincide with participant fractions, mean local ancestry, or meta-analysis weights.

The frequency channel, implemented in SMARTpred, recovers a known three-population mixture to within 0.01 in a unit-test panel after random allele swaps, and on a predeclared Yengo et al. (2022) height control set it ranks the corresponding 1000 Genomes super-population first in all five ancestry-stratified analyses (snapshot `yengo-height-2026-08-30`). The pair-product LD channel recovers a known two-population composition under a model-consistent null simulation (mean absolute error < 0.05) and ranks the declared reference first on the same five real studies, but its normalised simplex weights are not validated by a sign-flip diagnostic and every real fit sits on the simplex boundary. The bilinear χ^2 channel recovers composition when its own moment model is the data-generating process, and otherwise declines; it declined on all six real studies.

These results support the frequency channel as a deployed EAF-profile diagnostic. They do not license wiring either LD estimator into a production web service, nor reading any of the weights as a census of study participants.

Contents

1 Introduction

This report derives and compares working models for ancestry-related composition in a GWAS summary-statistics dataset. There is no universal "ancestry proportion" estimand. The frequency channel estimates the simplex projection of the reported effect-allele frequencies (EAF) onto reference-population frequencies. Under additional sampling assumptions, that projection may approximate allele-count or participant fractions. The LD channels estimate coefficients in a model for the covariance of GWAS z-statistics. Only under a restrictive discrete cohort-mixture model do those coefficients approximate the same source-population fractions. Neither target is a person's local-ancestry mosaic, and neither generally equals the variant-specific contribution of each study to a meta-analysis.

The frequency channel is implemented in SMARTpred and follows Summix [3, 13] and the bigsnpr ancestry resources [9, 12]. The LD family uses no allele frequencies: it compares products $z_i z_j$ or squared statistics χ_i^2 with per-population LD references. Z-score-based ancestry estimation was introduced within DISTMIX2 [5] and refined as the standalone ZMIX estimator [6]; the moment calculation is related to LD-score regression [4]. The original DISTMIX method [7] instead estimated weights from cohort allele frequencies (or accepted user-specified weights) before constructing an ancestry-informed LD matrix for summary-statistic imputation.

The practical motivation is threefold. First, some summary statistics ship without a usable frequency column, or with one that cannot be trusted; the LD statistics can be formed from $\hat{\beta}_i$ and SE_i, although valid interpretation also needs sample-overlap, test-construction and reference metadata. Second, the two channels interrogate different summaries, but they are not statistically independent: they can share participants, genotypes, reference error and selection. Agreement is therefore a useful descriptive diagnostic only after checking that they target comparable weights; disagreement is not by itself evidence that either channel is correct. Third, for cross-ancestry polygenic-score work such as the portability measurements of this repository, a calibrated compatibility check between each input and its declared reference population would be useful.

The evidential status is now empirical as well as formal. Section 13 records the encoded simulation benchmarks; Section 14 records the versioned Yengo et al. (2022) height snapshots. Estimator A remains experimental: it ranks the declared reference first on those five studies, but its normalised weights fail a sign-flip check and every real fit is on the simplex boundary. Estimator B remains a research prototype and declined on all six real analyses. Neither LD channel should be integrated into SMARTpred on the present evidence.

The report is organised as follows. Section 2 fixes notation. Section 3 states the frequency-based estimator. Sections 4 and 5 derive what the association statistics know about the analysed population's LD. Sections 6 and 7 define the two LD-moment estimators. Sections 8 – 10 treat uncertainty, identifiability, and failure modes; Section 11 positions the estimators against prior work. Section 12 maps them onto the code bases. Sections 13 and 14 report simulation and real-data benchmarks.

2 Setting and notation

Let $i = 1, \ldots, m$ index variants (in practice a HapMap 3 panel [2] of roughly one million variants, harmonised across references), and let K reference ancestries be available with LD correlation matrices $R^{(k)} = (\rho_{ij}^{(k)})$ on the standardized-genotype scale, estimated from ancestry-matched reference panels (for example 1000 Genomes super-populations [1]; in the ldpred3 ecosystem, UK Biobank references of the LDpred2 family [10]). The analysed GWAS dataset provides

$$z_i \;=\; \frac{\hat{\beta}_i}{\mathrm{SE}_i}, \qquad \chi_i^2 \;=\; z_i^2, \tag{1}$$

with effective sample size n (per-variant n_i where available), and optionally a reported effect-allele frequency f_i. A generic vector of non-negative reference weights lies on the simplex

$$w \;=\; (w_1, \ldots, w_K) \;\in\; \Delta_K \;=\; \Big\{ w : w_k \geq 0,\ \textstyle\sum_k w_k = 1 \Big\}. \tag{2}$$

The symbol must carry its estimand. We use π^{AF} for EAF projection weights and π^{LD} for coefficients in the LD working model. A participant fraction counts people assigned to each source; a genome-wide ancestry fraction averages inherited ancestry tracts; and local ancestry varies by locus and haplotype. Meta-analysis contributions are yet another object, because study and sample weights can vary by variant. These quantities coincide only in special designs and are not silently interchanged below.

3 The frequency channel (implemented)

Let $f = (f_1, \ldots, f_m)$ denote the *reported* EAF vector after allele harmonisation. The frequency-channel working model is

$$f_i \;\approx\; \sum_{k=1}^{K} \pi_k^{\mathrm{AF}}\, p_i^{(k)} \;+\; \varepsilon_i, \tag{3}$$

with $p_i^{(k)}$ the panel frequency of variant i's counted allele in ancestry k. The estimator minimises the residual sum of squares over the simplex,

$$\hat{\pi}^{\mathrm{AF}} \;=\; \operatorname*{arg\,min}_{w \in \Delta_K} \sum_{i=1}^{m} \Big(f_i - \sum_k w_k p_i^{(k)} \Big)^2, \tag{4}$$

after matching variants by identifier and allele pair (swapped alleles map f_i to $1 - f_i$; strand-ambiguous variants are excluded, since frequency-based strand resolution is unreliable exactly where ancestries differ). This is the Summix estimator [3] up to optimisation details, and it is what the SMARTpred ancestry mode implements with chromosome jackknife standard errors and panel-confusability diagnostics. Its requirement — a trustworthy EAF column with known provenance — is precisely what the channels of Sections 6–7 dispense with. Equation (4) estimates a geometric projection, not automatically a participant fraction. The stronger interpretation requires that f_i estimates the pooled allele frequency of a common sample, that the reference populations span that sample, and that drift, relatedness, participation or survival selection, case ascertainment and variant-specific missingness do not change the weights. A case-only EAF, a case–control average, or a meta-analytic EAF can target different and variant-specific mixtures.

4 What the association statistics know

Write the marginal association statistic of variant i as

$$z_i \;=\; \frac{1}{\sqrt{n}}\, x_i^{\top} y, \qquad y \;=\; \sum_{j=1}^{m} x_j \gamma_j \;+\; \varepsilon, \tag{5}$$

where x is the standardized genotype matrix of the *analysed* population A (mean zero, variance one per variant), the per-variant true standardized effects satisfy $\mathrm{Var}(\gamma_j) = h^2/m$ (infinitesimal model), $\mathrm{Var}(\varepsilon) = 1 - h^2$, and n is the effective sample size. Taking expectations over effects and noise, with $R^{(A)} = (\rho_{ij}^{(A)})$ the LD of the analysed population,

$$\mathbb{E}[z_i z_j] \;=\; \frac{nh^2}{m}\,(R^{(A)2})_{ij} \;+\; (1-h^2)\,\rho_{ij}^{(A)} \qquad \text{for all } i, j, \tag{6}$$

and, setting $i = j$ with $\rho_{ii}^{(A)} = 1$ and $\ell_i^{(A)} = \sum_j (\rho_{ij}^{(A)})^2$ the LD score of variant i in the analysed population,

$$\mathbb{E}\Big[\chi_i^2\Big] \;=\; 1 \;-\; h^2 \;+\; \frac{nh^2}{m}\,\ell_i^{(A)} \;\approx\; a + b\,\ell_i^{(A)}, \tag{7}$$

the LD-score-regression identity [4]: the intercept a absorbs $1-h^2$ plus any confounding inflation, and the slope $b = nh^2/m$ is the polygenic signal. Under the particular model in Equation (5), Equations (6)– (7) link second moments of the association statistics to the analysed sample's test-statistic covariance. Inferring source weights then requires the additional reference-mixture approximation in Section 5; the moment identities alone identify no ancestry quantity.

Two quantitative remarks. First, the noise term for a pair is proportional to $\rho_{ij}^{(A)}$, whereas the tagging term $(R^{(A)2})_{ij} = \sum_r \rho_{ir}^{(A)}\rho_{rj}^{(A)}$ accumulates shared LD paths and need not be proportional to the pair's direct correlation. Both terms vanish for pairs in independent LD blocks; arbitrary nominally "unlinked" pairs are not guaranteed to do so. Second, under the rough within-block heuristic $|(R^{(A)2})_{ij}| \approx \bar{\ell}|\rho_{ij}^{(A)}|$, the noise term dominates the tagging term by

$$\frac{(1-h^2)|\rho_{ij}^{(A)}|}{(nh^2/m)|(R^{(A)2})_{ij}|} \;\approx\; \frac{1-h^2}{nh^2\bar{\ell}/m} \;\approx\; \frac{0.75}{0.05} \quad \text{at } n = 10^5,\ h^2 = 0.25,\ m = 10^6,\ \bar{\ell} \approx 2: \tag{8}$$

the pair-product moment is driven by correlated estimation noise under Equation (5), including at negligible polygenic signal. Correct marginal standard errors are necessary but not sufficient for arbitrary GWAS test constructions to have this covariance. The χ^2 moment, by contrast, requires genuine signal ($h^2 > 0$), since at $h^2 = 0$ the expectation in (7) is flat in ℓ.

5 Mixture model for the analysed population's LD

The linear LD mixture is a working model, not a generic identity for a pooled or admixed cohort. If S denotes a discrete source label with participant fractions q_k, the law of total covariance gives

$$\mathrm{Cov}(G_i, G_j) = \sum_k q_k\,\mathrm{Cov}(G_i, G_j \mid S = k) + \sum_k q_k(\mu_{ik} - \mu_i)(\mu_{jk} - \mu_j), \tag{9}$$

where $\mu_{ik} = \mathbb{E}(G_i \mid S = k)$. The second term is the between-source or Wahlund component; conversion from covariance to correlation also introduces source- and mixture-specific genotype variances. Recent admixture adds covariance between local-ancestry tracts. Consequently participant fractions, mean local-ancestry fractions and LD mixture coefficients need not agree.

For linked pairs, suppose instead that the GWAS test covariance is well approximated by a convex combination of the harmonised reference correlations. Define π^{LD} as the coefficients of that approximation,

$$\rho_{ij}^{(A)} \;\approx\; \sum_{k=1}^{K} \pi_k^{\mathrm{LD}}\,\rho_{ij}^{(k)} \qquad \text{for linked } (i, j). \tag{10}$$

This approximation is most plausible for a discrete mixture of source-group participants analysed with a common, unadjusted score-test construction, after within-source standardisation, and when the extra term in (9) is negligible or modelled. It is not a claim that π^{LD} equals an individual-level admixture fraction.

Substituting (10) into the noise term of (6) keeps that component linear in π^{LD}. The χ^2 route is different: the LD score squares the correlation, so

$$\ell_i^{(A)} = \sum_j \Big(\sum_k \pi_k^{\mathrm{LD}} \rho_{ij}^{(k)}\Big)^2 = \sum_k (\pi_k^{\mathrm{LD}})^2 \ell_i^{(kk)} + 2 \sum_{k<k'} \pi_k^{\mathrm{LD}} \pi_{k'}^{\mathrm{LD}} \ell_i^{(kk')}, \qquad \ell_i^{(kk')} := \sum_j \rho_{ij}^{(k)} \rho_{ij}^{(k')}, \tag{11}$$

which is linear in products of the coefficients, not in the coefficients themselves. The factor two in the off-diagonal terms is essential when only the $K(K+1)/2$ unique bilinear scores are used. These scores are computable once from each unordered pair of reference panels and form the design in Section 7. Note that regressing χ^2 on the plain per-ancestry scores $\ell_i^{(k)} = \ell_i^{(kk)}$ instead is misspecified whenever the cross-ancestry correlation vectors overlap, as they generally do in real panels; the cross terms in Equation (11) are what that shortcut drops.

6 Estimator A: pair products (linear LD channel)

For a set P of sampled variant pairs with non-trivial reference LD, use the noise term of Equations (6) and (10). When the polygenic tagging term is negligible, the raw working model is

$$\mathbb{E}[z_i z_j] \approx s_0 \sum_{k=1}^{K} \pi_k^{\mathrm{LD}} \rho_{ij}^{(k)}, \qquad (i,j) \in P. \tag{12}$$

At truth $s_0 = 1 - h^2$. A single scale cannot in general absorb the tagging term $(nh^2/m)(R^{(A)2})_{ij}$: $R^{(A)2}$ is not proportional to $R^{(A)}$. Expanding the square of the LD mixture gives

$$R^{(A)2} \approx \sum_k (\pi_k^{\mathrm{LD}})^2 R^{(k)} R^{(k)} + \sum_{k<k'} 2\pi_k^{\mathrm{LD}} \pi_{k'}^{\mathrm{LD}} Q^{(kk')}. \tag{13}$$

For $k \le k'$, define the quadratic absorber matrices

$$Q^{(kk)} = R^{(k)} R^{(k)}, \qquad Q^{(kk')} = \frac{R^{(k)} R^{(k')} + R^{(k')} R^{(k)}}{2} \quad (k < k'). \tag{14}$$

The order of the two cross-products matters because symmetric LD matrices need not commute. Using only $R^{(k)} R^{(k')}$ for $k < k'$ therefore does not span the tagging term exactly. An absorbed linear fit uses

$$z_i z_j \approx \sum_k a_k \rho_{ij}^{(k)} + \sum_{k \le k'} c_{kk'} Q_{ij}^{(kk')}, \qquad (i,j) \in P, \tag{15}$$

with non-negative coefficients. Under the full working model, write $q = nh^2/m$. Then $a_k = (1-h^2)\pi_k^{\mathrm{LD}}$, $c_{kk} = q(\pi_k^{\mathrm{LD}})^2$, and $c_{kk'} = 2q\pi_k^{\mathrm{LD}} \pi_{k'}^{\mathrm{LD}}$ for $k < k'$. The composition estimate comes from the linear coefficients,

$$\hat{\pi}_k^{\mathrm{LD}} = \frac{\hat{a}_k}{\sum_{r=1}^{K} \hat{a}_r}. \tag{16}$$

Equation (16) is defined only when $\sum_r \hat{a}_r > 0$. If the linear component vanishes, Estimator A declines; the row sums of the quadratic nuisance coefficients are not substituted as the target composition. Treating the $c_{kk'}$ as nuisance coefficients protects the linear channel from tagging-term bias, but does not repair a false LD-mixture model and can worsen collinearity.

The implementation retains these absorbers only when their fitted total exceeds twice a naive parametric standard error. Because pair residuals are LD-correlated, this is an uncalibrated activation heuristic, not evidence of "real signal"; a reported zero signal means only that the absorber was not retained. Requirements and properties are therefore:

1. **No EAF column and no known h^2.** At $h^2 = 0$, the correlated-noise term remains under the eligible score-test model. Conversely, the linear channel weakens as residual variance $1-h^2$ approaches zero. This statement does not extend automatically to arbitrary adjusted, case–control or meta-analytic z-statistics.
2. **Pairs must be linked.** Unlinked pairs have $\rho_{ij}^{(k)} \approx 0$ for every k and contribute only noise; take pairs within LD blocks, e.g. all pairs with $\max_k |\rho_{ij}^{(k)}|$ above a floor, capped per block to bound cost.
3. **Genome-wide z's.** Because each pair is weakly informative, the estimator needs statistics across all chromosomes; a single region does not suffice.
4. **The linear design must be identifiable after conditioning on the absorbers.** Boundary solutions and nearly collinear reference columns are failure diagnostics, not precise ancestry calls.

The raw pair-product idea is the estimator family of ZMIX [6], following its use inside DISTMIX2 [5]. The quadratic absorber is an extension studied here. The original DISTMIX [7] is not a Z-score-based ancestry estimator.

7 Estimator B: bilinear LD-score regression

For a common sample size, the uncapped moment implied by Equations (7) and (11) is

$$\mathbb{E}\Big[\chi_i^2\Big] \;=\; a + \sum_k b_{kk}\ell_i^{(kk)} + \sum_{k<k'} b_{kk'}\ell_i^{(kk')}. \tag{17}$$

With $s = nh^2/m$, the population coefficients under the working model are

$$b_{kk} = s(\pi_k^{\mathrm{LD}})^2, \qquad b_{kk'} = 2s\pi_k^{\mathrm{LD}}\pi_{k'}^{\mathrm{LD}} \quad (k < k'). \tag{18}$$

Define a symmetric matrix B by $B_{kk} = b_{kk}$ and $B_{kk'} = B_{k'k} = b_{kk'}/2$ for $k < k'$. At truth, $B = s\pi^{\mathrm{LD}}\pi^{\mathrm{LD}\top}$, so the composition can be recovered from its normalised row sums:

$$\hat{\pi}_k^{\mathrm{LD}} \;=\; \frac{\sum_{k'} \hat{B}_{kk'}}{\sum_k \sum_{k'} \hat{B}_{kk'}}. \tag{19}$$

The scale s is not separately needed for Equation (19), but it must be positive and estimable: at $h^2 = 0$ the composition is unidentified. Non-negativity alone is insufficient. The fitted design must have adequate rank and conditioning, and $\hat{B}$ must be close to positive semidefinite and rank one; otherwise row-sum normalisation can manufacture a composition from a misspecified coefficient pattern. For $K = 4$ there are $K(K+1)/2 = 10$ signal columns, or 11 regression columns including the intercept.

The implementation uses the uncapped response by default. A caller may request $\min(\chi_i^2, c)$ as an exploratory robustness diagnostic, but if any observation is actually truncated the implementation withholds the composition. This is necessary because $\mathbb{E}\{\min(\chi_i^2, c)\}$ is nonlinear in

$\mathbb{E}(\chi_i^2)$. Filtering exceptional loci, a two-stage LDSC-style fit, or a robust loss should be compared and calibrated before any such fit is used inferentially.

The current acceptance screen requires the centred bilinear design to have full rank and scale-normalised condition number at most 1000; at least four usable held-out block groups; a mean held-out correlation divided by its between-group standard error of at least 4; relative rank-one distance at most 0.25; negative-eigenvalue magnitude at most 0.10 of the fitted matrix's spectral scale; and no actual response truncation. These are fixed engineering thresholds, not calibrated hypothesis tests.

Per-variant sample sizes enter the signal multiplicatively:

$$\mathbb{E}(\chi_i^2) \approx a + \frac{h^2}{m} n_i \ell_i^{(A)}. \tag{20}$$

Thus the signal design uses $n_i \ell_i^{(kk')}$, not $\ell_i^{(kk')}/n_i$. Dividing the response by n_i is not equivalent unless the resulting variant-specific intercept is modelled. Estimator B is a cheap per-variant regression after precomputation, but its stronger signal and specification requirements leave it a research prototype.

8 Uncertainty

Residuals of both estimators are correlated across nearby variants and pairs through LD, so i.i.d. standard-error formulas are wrong. Use the delete-one-chromosome (or delete-one-LD-block) jackknife: refit on all but one chromosome's variants (pairs), and form

$$\mathrm{SE}(\hat{\theta}_k) \;=\; \sqrt{\frac{g-1}{g} \sum_{g'=1}^{g} \left(\hat{\theta}_k^{(-g')} - \bar{\hat{\theta}}_k\right)^2}, \tag{21}$$

where θ denotes the channel-specific target and g is the number of informative groups; the same construction the frequency channel uses. Chromosomes contain unequal numbers of informative variants and pairs, so the equal-group formula in Equation (21) is approximate; approximately equal-information LD blocks or a weighted delete-group jackknife should be included in calibration studies. For Estimator A, $\mathrm{Var}(z_i z_j) \approx 1 + \{\mathbb{E}[z_i z_j]\}^2$ under approximate normality suggests weights $1/(1 + \hat{\rho}^2_{\mathrm{fit},ij})$; the unweighted fit with jackknife is the simpler default. Estimator B reports a delete-group SE only when every leave-group refit passes the same design, held-out-signal, positive- semidefinite and rank-one screens as the full fit; otherwise the SE is withheld. This is still only block-aware uncertainty under its uncapped default, not calibration of the fixed acceptance thresholds. Exploratory capped or robust alternatives require separate uncertainty work.

Simplex boundaries are non-regular. A point estimate of zero does not prove absence of that source, symmetric normal intervals can leave the simplex, and delete-group replicates may pile up at zero. Boundary-aware confidence sets or calibrated one-sided limits are preferable when a scientific claim depends on a small component.

9 Identifiability

The simplex constraint identifies frequency-projection weights only if the reference-frequency contrasts have adequate rank on the matched variants. Estimator A identifies the relative linear coefficients a_k in Equation (15); its nuisance absorber can remove directions that would otherwise distinguish sources. Estimator B identifies a composition only through a non-zero, scaled rank-one matrix. Under that model its overall scale is $s = \mathbf{1}^\top B\mathbf{1}$ and is algebraically separable from the normalised composition; this does not by itself validate s as the biological quantity nh^2/m. The composition is undefined when $s = 0$.

The deeper constraint is geometric. The ancestry design columns (frequencies $p^{(k)}$, pair correlations $\rho_{ij}^{(k)}$, or bilinear scores $\ell^{(kk')}$) can be severely collinear. LD patterns in particular often decay with distance similarly across sources, so the LD design can be worse conditioned than the frequency design. Rank and condition numbers must be computed on the *matched, fitted contrast design*, after including nuisance columns, rather than on the full unfiltered panel. Pairwise-column correlations, effective rank, boundary status and, for Estimator B, distance from the positive-semidefinite rank-one cone should be reported. Near-collinear sources should be pooled or the analysis should decline; returning an arbitrary simplex vertex is not identification.

10 Assumptions and failure modes

Assumption 1 (Eligible LD-mixture design)**.** *The approximation in Equation* (10) *holds for the covariance of the GWAS test statistics on the fitted linked pairs. This is an assumption about the analysed cohort and its test construction, not merely about the reference panels. Equation* (9) *shows the omitted between-source term. Admixture LD adds long-range covariance proportional to ancestry-frequency contrasts and local-ancestry correlation; recent admixture is the hard case. In these settings π^{LD} is at most a best-fitting reference coefficient, not an identified participant or local-ancestry fraction.*

Assumption 2 (Infinitesimal effects)**.** $\mathrm{Var}(\gamma_j) = h^2/m$ *with exchangeable effects on the admixed-genotype scale. A trait driven by a few large loci violates the moment structure of* (6)–(18)*; the uncapped default can be dominated by such loci. Opt-in response capping limits influence but changes the expectation and causes the implementation to withhold proportions. It is not a general correction for sparse or source-heterogeneous genetic architecture.*

Assumption 3 (Honest standard errors)**.** *The dominant term of Estimator A is the correlated-noise covariance: miscalibrated, winsorised, or meta-analysis-heterogeneity-distorted SEs contaminate it directly. This is the channel's greatest sensitivity.*

Assumption 4 (Compatible GWAS adjustment)**.** *Stratification and uncontrolled confounding add LD-score-shaped and pair-product-shaped components (the LDSC intercept story of Bulik-Sullivan et al. [4]). Estimator B's free intercept absorbs only their mean component; Estimator A has no intercept. Conversely, adjustment for principal components, other genetic covariates, or a mixed model changes the score covariance. If M_C residualises the GWAS covariates, the relevant reference object is proportional to*

$$R_C = D_C^{-1/2} X^\top M_C X D_C^{-1/2}, \qquad D_C = \mathrm{diag}(X^\top M_C X), \tag{22}$$

not generally the raw-genotype LD R. Covariate-adjusted LDSC likewise uses adjusted, preferably in-sample LD [8]. Without matching the target test construction, either estimator can mistake adjustment or confounding for source composition.

Assumption 5 (Stable participants and overlap)**.** *Equations* (6)–(7) *assume a common participant set. With variant-specific missingness, the noise covariance for a pair is attenuated by its participant overlap, approximately $n_{ij}/\sqrt{n_i n_j}$ under simple sampling. A single per-variant n_i*

does not determine n_{ij}. Unknown overlap patterns can therefore mimic an LD difference and invalidate both the fit and its jackknife.

Assumption 6 (Single-cohort quantitative-trait score model)**.** *The derivation uses marginal linear scores for a standardised quantitative trait. Logistic case–control statistics depend on ascertainment, case fraction and covariate adjustment; their accompanying EAF may describe cases, controls or a pooled sample. Meta-analysis statistics combine study-specific scores with variant-specific weights, sample overlap and heterogeneity. In either setting, a single constant π^{LD} or π^{AF} requires an explicit derivation for that construction. Otherwise the estimator should decline rather than silently attach a cohort-fraction interpretation. Selection into participation, survival or analysis can also change EAF and LD relative to both the source population and the external panel; the result then describes the selected analysed sample at best.*

Assumption 7 (Harmonised references)**.** *All K references are built on the same variant panel with consistent allele orientation, so pair correlations and bilinear scores are comparable across ancestries. Panel construction from 1000 Genomes [1] or UK Biobank references [10] must pass through the same harmonisation as the target statistics. Finite reference samples add correlated measurement error to every design column. A target that includes an unrepresented source is projected onto the nearest available convex combination. Raw EAF correlation can remain high because all populations share a large baseline, so it does not rule out such misspecification; goodness-of-fit should be evaluated in contrast space. Reference-panel resampling, held-out variants, contrast-space goodness-of-fit and explicit out-of-panel simulations are needed. The GWAS block jackknife alone does not include reference uncertainty.*

11 Relation to prior work

Table 1 summarises the channels. *Summix* [3] and its successor [13] deconvolve ancestry from allele frequencies with a mixture model, and adjust frequencies to a target composition; the bigsnpr ancestry resources [9, 12] implement the same idea through projection onto reference PCs followed by simplex-constrained quadratic programming, with a worldwide 21-group reference. *DISTMIX* [7] infers mixing proportions from cohort allele frequencies, or accepts prespecified weights, in order to build ancestry-informed LD matrices for imputing summary statistics. *DISTMIX2* [5] introduced a Z-score-only weight estimator within an expanded direct-imputation workflow. *ZMIX* [6] made that task standalone and uses z-score products with constrained optimisation; it reports less accurate estimation for the z channel than for frequency-based methods, especially for admixed targets. Estimator A is in this family, with the symmetrised quadratic absorber in Equation (14) an extension evaluated here.

The misspecification checks of Privé et al. [11] ask whether a reference is compatible with summary statistics. Composition estimation is a stronger inverse problem: it additionally assumes that incompatibility can be represented by the supplied convex mixture. Estimator B's bilinear scores are, to the best of our knowledge, a new exploratory ancestry estimator related to LD-score regression [4]. That is not a priority claim over methods not surveyed exhaustively, nor evidence of identification in real GWAS designs.

12 Implementation mapping and recommendation

In SMARTpred, the frequency channel is implemented in `smartpred/ancestry.py` with registered, hash-pinned 1000 Genomes panels, chromosome-jackknife standard errors and confusability reporting. The Yengo controls in Section 14 are the present real-data positive-control record

Table 1: Ancestry-composition channels from summary statistics.

Channel	Data used	Moment	Prior work
Frequency	EAF column f_i	first, (3)	Summix [3], [13], bigsnpr [12]
Pair products	$z_i z_j$, linked pairs	second, (12)	DISTMIX2 [5], ZMIX [6]
Bilinear LD scores	χ_i^2, all variants	second, (17)	LDSC generalisation [4]; new here

for that channel. They do not replace out-of-panel negative controls or a case/control EAF provenance study.

Estimator A's block-native implementation uses the Jordan-product absorber in Equation (14); refuses rank-deficient or ill-conditioned raw and absorber-conditioned linear designs; and reports block-jackknife uncertainty. The same Yengo snapshot shows that ranking the declared reference first is not the same as validating the simplex weights. Estimator B should remain a research prototype: its present signal, design, positive-semidefinite and rank-one screens are heuristic tolerances, and they declined on every real study in that snapshot. Neither LD estimator should be wired into SMARTpred on this evidence.

For this repository, a validated estimator could eventually serve as a preflight compatibility check in the cross-ancestry portability pipeline. It could flag a dataset whose summary-statistic covariance is incompatible with its declared reference. It cannot by itself verify self-identified ancestry, infer individual local ancestry, or prove a particular participant mixture. Those claims require participant-level labels or an independently validated design.

13 Simulation experiments

The numerical claims below are the encoded simulation gates in `tests/test_ancestry_frequency.py` and `tests/test_ancestry.py`. They are not a Monte Carlo survey of operating characteristics. A fuller LD-moment grid lives in `experiments/ancestry_ld_study.py`; that script is a study harness, not a results registry.

Frequency channel. On a synthetic three-population panel of $m = 3000$ variants, a known mixture $\pi^{\mathrm{AF}} = (0.60, 0.25, 0.15)$ plus independent $N(0, 0.002^2)$ frequency noise is recovered to within 0.01 in absolute error after a 30% random allele swap that the matcher is required to reverse. The returned status is `estimated`, the observed-versus-fitted frequency correlation exceeds 0.99, and the reported reference-sampling RMS matches the binomial formula evaluated at the fitted weights. Inputs with an out-of-range EAF are `invalid_input`. Panels with fewer than 1,000 matched variants, fewer than ten autosomes, or more than 25% of matches on one chromosome are `insufficient`. A leave-one-chromosome contrast that loses rank is refused.

Pair-product channel (Estimator A). Under model-consistent multivariate-normal z-draws with true composition $\pi^{\mathrm{LD}} = (0.65, 0.35)$, $h^2 = 0$ and $n = 1000$, eight replicates on a 400-block landscape recover π with mean absolute error below 0.05, and the chromosome-jackknife standard errors lie within a factor of two of the empirical replicate standard deviation. At the tagging-dominated regime $nh^2/m = 0.5$, the raw (unabsorbed) fit has mean absolute error above 0.25 and saturates on the high-LD ancestry; retaining the quadratic absorber in Equation (14) reduces the error below 0.20 on the accepted replicates (at least nine of twelve draws; the rest decline because the linear component is not positive). At the report's more realistic $nh^2/m = 0.025$, both raw and absorbed fits have mean absolute error below 0.14. A four-population composition

$(0.4, 0.3, 0.2, 0.1)$ is recovered with mean absolute error below 0.10. When the linear component vanishes, the fit declines rather than substituting quadratic nuisance coefficients as the composition.

On individual-level mosaic genotypes (not the moment model), replicate-averaged recovery of $(0.65, 0.35)$ has mean absolute error below 0.15 at $F_{\mathrm{ST}} = 0.05$ and below 0.17 at $F_{\mathrm{ST}} = 0.2$. Boundary estimates, when they occur, carry a warning rather than a confident zero standard error.

Bilinear channel (Estimator B). When χ^2 statistics are generated from the bilinear scores themselves plus small Gaussian noise, Estimator B recovers $(0.65, 0.35)$ to within 0.01, with relative rank-one distance below 0.01 and no simplex-boundary note. At $h^2 = 0$ on model-consistent z-draws it declines for insufficient reproducible held-out signal, including on ten seeds that a former i.i.d. two-standard-error guard had accepted. Duplicate or near-duplicate references are reported as non-identifiable rather than as a simplex vertex. On mosaic GWAS draws with $h^2 = 0$ it likewise declines and reports a reason.

14 Real-data experiments

The real-data record is the pair of versioned snapshots of 30 August 2026 (frequency and LD). Both use the ancestry-stratified height GWAS of Yengo et al. [14]: five super-population-labelled analyses plus a pooled meta-analysis, all of the same trait and publication. The five labelled analyses are predeclared qualitative controls. The pooled analysis is descriptive and is excluded from the verdict. Numeric acceptance gates were not tuned on these files.

The frequency panel is the hash-pinned 1000 Genomes phase-3 super-population ALT-frequency panel at HapMap 3 sites (1,054,330 non-palindromic GRCh37 variants; semantic SHA-256 `c91b6652...b7291186`; sample counts 661/347/504/503/489 for AFR/AMR/EAS/EUR/SAS). The LD design is a compact, target-blind subset of the same panel: 714,078 common HM3 variants, 177 blocks separated by at least 5 Mb, 21,892 variants and 44,250 pairs. LD is estimated in the published 2,490-sample mostly-unrelated 1000 Genomes subset. Pair selection and LD estimation reuse those samples, so reference winner's-curse and errors-in-variables uncertainty are absent from the chromosome jackknife.

Frequency channel. Table 2 reports the EAF-projection weights. All five controls returned status `estimated` and ranked the predeclared 1000 Genomes super-population first (5/5). Matched variant counts are about 1.04 million. Observed-versus-fitted frequency correlations are 0.994–0.999. Residual RMS values lie between 0.012 and 0.033, below the engineering gate of 0.05. The Hispanic/Latin American analysis is a predeclared imperfect proxy for 1000 Genomes AMR; it still ranked AMR first, with weight 0.819. The European analysis is nearly a vertex (0.989 EUR). The African analysis retains a non-negligible EUR component (0.132), which is a statement about deposited EAF relative to this five-population basis, not a census of participants. The pooled meta-analysis is EUR-dominant (0.766) with a visible EAS component (0.112), as expected from the sample-size mix, and is not a control.

Pair-product channel. Table 3 repeats the Estimator A weights from the LD snapshot. The declared reference ranked first in all five controls (5/5). AMR sat at a vertex. Every fit, including the pooled analysis, touched a simplex boundary, so ordinary symmetric jackknife intervals are not a valid confidence statement; AMR's reported zero SE is a boundary artifact. A scaled sign-flip diagnostic of the form $s(\pi_{\text{expected}} - \max_{k \neq \text{expected}} \pi_k)$ was never exceeded by any of 200 sign-flip replicates per study (plus-one $p = 1/201 = 0.0050$ on the permissive reading that treats a declined replicate as a non-exceedance). The corresponding normalised-weight contrasts did not clear $p \leq 0.05$. Scaled correct-reference direction is therefore supported; normalised mixture magnitudes are not. Estimator B declined on all six studies for insufficient reproducible LD-dependent χ^2 signal. Sparse marker pruning also omits genome-wide tagging paths, so this is not a fair genome-scale verdict on B.

Table 2: EAF-projection weights on the Yengo et al. (2022) height analyses. Entries are $\hat{\pi}^{\text{AF}}$ (delete-one-chromosome jackknife SE). The five labelled studies are predeclared controls; the pooled analysis is excluded from the 5/5 verdict. Snapshot `yengo-height-2026-08-30`.

Study	N	AFR	AMR	EAS	EUR	SAS	r
AFR	168 193	0.835 (0.001)	0.017 (0.001)	0.007 (0.000)	0.132 (0.001)	0.009 (0.001)	0.999
AMR	58 709	0.040 (0.001)	0.819 (0.004)	0.025 (0.001)	0.103 (0.003)	0.014 (0.001)	0.997
EAS	363 856	0.000 (0.000)	0.030 (0.001)	0.970 (0.001)	0.000 (0.000)	0.000 (0.000)	0.994
EUR	1 597 374	0.000 (0.000)	0.008 (0.001)	0.000 (0.000)	0.989 (0.001)	0.004 (0.001)	0.998
SAS	60 939	0.000 (0.000)	0.023 (0.001)	0.000 (0.000)	0.176 (0.001)	0.801 (0.001)	0.998
Pooled*	2 200 007	0.046 (0.001)	0.047 (0.001)	0.112 (0.002)	0.766 (0.002)	0.030 (0.001)	0.998

N is the reported GWAS sample size. r is the observed-versus-fitted EAF correlation. *Descriptive; not a control. Jackknife SEs condition on the fixed panel.

Table 3: Estimator A LD-mixture weights on the same Yengo analyses. All five controls ranked the declared reference first. Every fit is on the simplex boundary. Snapshot `yengo-height-2026-08-30`.

Study	AFR	AMR	EAS	EUR	SAS	Boundary
AFR	0.702	0.170	0.000	0.113	0.014	yes
AMR	0.000	1.000	0.000	0.000	0.000	yes
EAS	0.000	0.000	0.961	0.021	0.017	yes
EUR	0.000	0.045	0.000	0.955	0.000	yes
SAS	0.000	0.004	0.000	0.110	0.887	yes
Pooled*	0.000	0.102	0.000	0.898	0.000	yes

*Descriptive; not a control. Estimator B declined on all six studies.

What would change this reading is a well-calibrated out-of-panel negative control, a case-only versus control-only EAF contrast on the same study, or a sign-flip (or other exchangeability) diagnostic that validated the *normalised* LD weights rather than only their scaled direction.

Acknowledgements

The derivations in Sections 4–7 were worked out in discussion of the SMARTpred ancestry mode; the frequency channel's implementation there served as the numerical template. The simulation gates are those of `tests/test_ancestry.py` and `tests/test_ancestry_frequency.py`. The real-data numbers are the committed snapshots of 30 August 2026. All citations were verified against the published records; any error in them is ours.

References

[1] Auton, A., Brooks, L. D., Durbin, R. M., et al. (1000 Genomes Project Consortium) (2015). A global reference for human genetic variation. *Nature*, 526(7571):68–74.

[2] Altshuler, D. M., Gibbs, R. A., Peltonen, L., et al. (International HapMap 3 Consortium) (2010). Integrating common and rare genetic variation in diverse human populations. *Nature*, 467(7311):52–58.

[3] Arriaga-MacKenzie, I. S., Mazee, A. M., Nguyen, A. P., et al., and Hendricks, A. E. (2021). Summix: A method for detecting and adjusting for population structure in genetic summary data. *American Journal of Human Genetics*. https://www.cell.com/ajhg/fulltext/S0002-9297(21)00221-4

[4] Bulik-Sullivan, B. K., Loh, P.-R., Finucane, H. K., et al. (2015). LD Score regression distinguishes confounding from polygenicity in genome-wide association studies. *Nature Genetics*, 47(3):291–295.

[5] Chatzinakos, C., Lee, D., Cai, N., et al. (2021). Increasing the resolution and precision of psychiatric genome-wide association studies by re-imputing summary statistics using a large, diverse reference panel. *American Journal of Medical Genetics Part B: Neuropsychiatric Genetics*, 186(1):16–27. https://doi.org/10.1002/ajmg.b.32834

[6] Dennis, T. and Lee, D. (2024). ZMIX: estimating ancestry proportions using GWAS association Z-scores. *Bioinformatics Advances*, 4(1):vbae128.

[7] Lee, D., Bigdeli, T. B., Riley, B. P., Fanous, A. H., and Bacanu, S.-A. (2015). DISTMIX: direct imputation of summary statistics for unmeasured SNPs from mixed ethnicity cohorts. *Bioinformatics*, 31(19):3099–3105.

[8] Luo, Y., Li, X., Wang, X., et al. (2021). Estimating heritability and its enrichment in tissue-specific gene sets in admixed populations. *Human Molecular Genetics*, 30(16):1521–1534. https://doi.org/10.1093/hmg/ddab130

[9] Privé, F., Luu, K., Blum, M. G. B., McGrath, J. J., and Visscher, P. M. (2018). Efficient analysis of large-scale genome-wide data with two R packages: bigstatsr and bigsnpr. *Bioinformatics*, 34(16):2781–2787.

[10] Privé, F., Arbel, J., Vilhjálmsson, B. J., et al. (2020). LDpred2: better, faster, and more accurate polygenic risk prediction. *American Journal of Human Genetics*, 107(6):1011–1021.

[11] Privé, F., Arbel, J., Aschard, H., and Vilhjálmsson, B. J. (2022). Identifying and correcting for misspecifications in GWAS summary statistics and polygenic scores. *Human Genetics and Genomics Advances*, 3(4):100136.

[12] Privé, F. (2022). *bigsnpr vignette: Ancestry — estimating ancestry proportions from allele frequencies* (software documentation). https://privefl.github.io/bigsnpr/articles/ancestry.html

[13] Stoneman, K. P., Arriaga-MacKenzie, I. S., Vasquez, K. G., et al., and Hendricks, A. E. (2025). Characterizing substructure via mixture modeling in large-scale genetic summary data. *American Journal of Human Genetics*. https://www.cell.com/ajhg/fulltext/S0002-9297(24)00449-X

[14] Yengo, L., Vedantam, S., Marouli, E., et al. (2022). A saturated map of common genetic variants associated with human height. *Nature*, 610(7933):704–712. https://doi.org/10.1038/s41586-022-05275-y